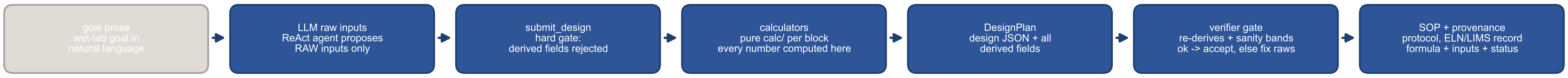
(a) pipeline topology — the boundary rule: the LLM proposes RAW inputs; deterministic calculators compute every number; the verifier re-proves



(b) field-level provenance DAG — built from provenance_for() on the plan above; every edge is a real input->field dependency, edges between derived fields carry the value that flows

flat one-step blocks (same provenance machinery, single level): flow → shear_pa · reynolds · pressure_drop_pa · residence_time_s · channel_volume_ul · mean_velocity_mms dosing → dmso_fraction_vv stats → n_per_group champ → n_arrays · n_chips

(c) conservation audit — additive identities recomputed from the same plan (the verifier's arithmetic layer re-proves exactly these)

cells N_A + N_B = 1.6e+07 vs ρ·A·wells = 1.6e+07 ✓ in = out

seed N_well × wells = 1.7e+07 vs seed_per_well × 6 = 1.7e+07 ✓ in = out

plan status: review_required (0 errors · 4 warnings) — warnings are the DMSO-safety and stats-n layers, intentionally kept so the status colouring is real